Practical no. 9B

Aim: Develop an E-Commerce PWA with product listing and offline cart functionality.

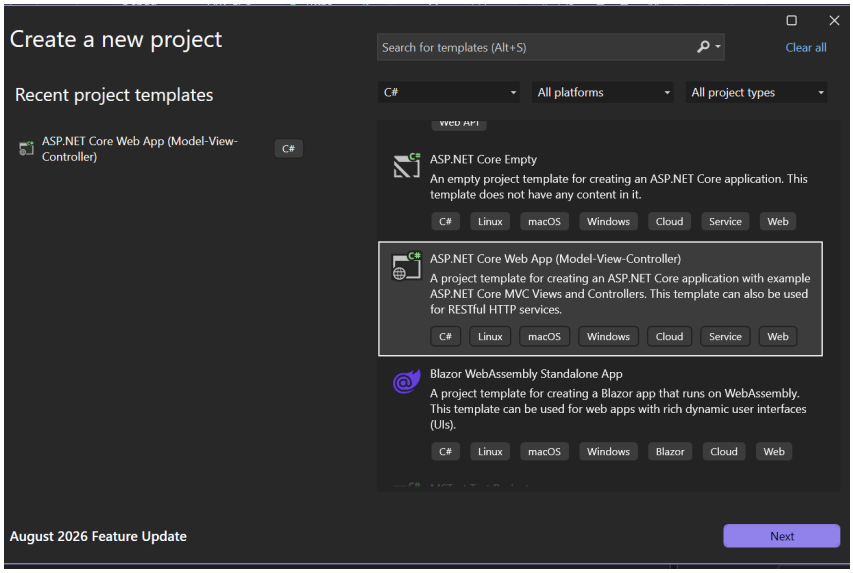
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Click: Create a New Project
3. Open Visual Studio 2026.
4. Select Create a new project.
5. Search for ASP.NET Core Web App (Model-View-Controller).
6. Click Next.
7. Set the Project Name to Practical No. 9A.
8. Select a suitable location and click Next.
9. Select the installed/latest .NET version and keep the default options.
10. Click Create.



STEP 2: CREATE HOME PAGE

Open Views → Home → Index.cshtml and replace its contents with:

```
@{
```

```
    ViewData["Title"] = "E-Commerce PWA";
}



<h1>E-Commerce Store</h1>

<hr>

<h2>Products</h2>

<div>

    <h3>Laptop</h3>
    <p>Price: ₹50000</p>

    <button onclick="addToCart('Laptop', 50000)">
        Add to Cart
    </button>

</div>

<hr>

<div>

    <h3>Mobile Phone</h3>
    <p>Price: ₹20000</p>

    <button onclick="addToCart('Mobile Phone', 20000)">
        Add to Cart
    </button>

</div>

<hr>

<div>

    <h3>Headphones</h3>
    <p>Price: ₹2000</p>

    <button onclick="addToCart('Headphones', 2000)">
        Add to Cart
```

```
    </button>

</div>

<hr>

<h2>Shopping Cart</h2>

<div id="cart">
</div>

<h3 id="total">
</h3>


<script src="/js/site.js"></script>
```

Step 3: Create site.js file
Open:

wwwroot → js → site.js

Add:

```
let cart = JSON.parse(localStorage.getItem("cart")) || [];

function addToCart(name, price) {

    cart.push({
        name: name,
        price: price
    });

    localStorage.setItem("cart", JSON.stringify(cart));

    displayCart();
}

function displayCart() {

    let cartDiv = document.getElementById("cart");
    let total = 0;

    cartDiv.innerHTML = "";
```

```
    cart.forEach(function (product) {

        cartDiv.innerHTML +=
            product.name + " - ₹" + product.price + "<br>";

        total = total + product.price;
    });

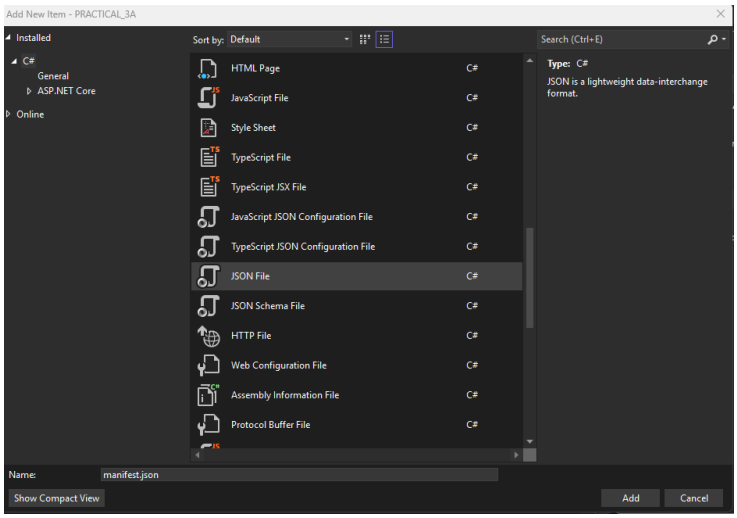
    document.getElementById("total").innerHTML =
        "Total: ₹" + total;
}

displayCart();

if ("serviceWorker" in navigator) {
    navigator.serviceWorker.register("/service-worker.js");
}
```

Step 4: Create Manifest File

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JSON file.
4. Enter File Name: **manifest.json**
5. Click Add.



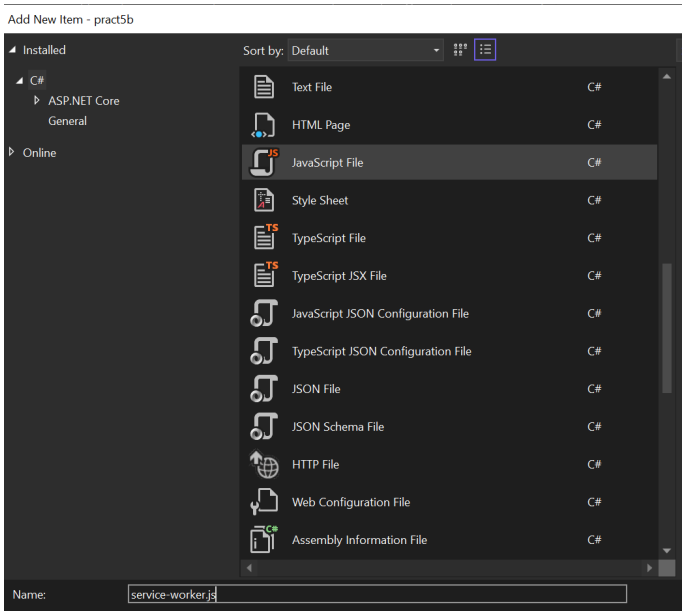
Write the following code in the manifest.json file.

manifest.json

```
{
 "name": "E-Commerce Store",
 "short_name": "E-Store",
 "start_url": "/",
 "display": "standalone"
}
```

Step 5: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JavaScript file.
4. Enter File Name: **service-worker.js**
5. Click Add.



Write the following code in the service-worker.js file.

service-worker.js

```
let cacheName = "shop-cache";

self.addEventListener("install", function (event) {

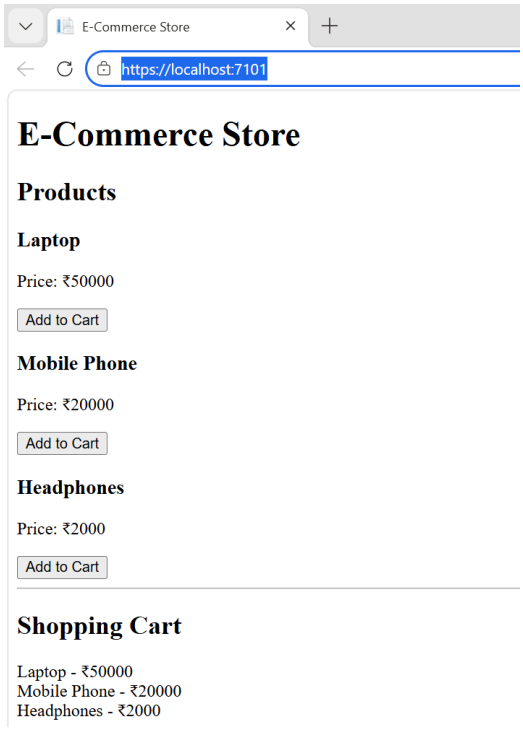
    event.waitUntil(

        caches.open(cacheName).then(function (cache) {

            return cache.add("/");
```

```
        })
    );
});
self.addEventListener("fetch", function (event) {
    event.respondWith(
        caches.match(event.request).then(function (response) {
            return response || fetch(event.request);
        })
    );
});
```

Step 6: Run the application
Press Ctrl + F5



Step 7: Test Offline Functionality
Open F12 → Network. → Select the Offline option and refresh the page.

